

$se_2(1)$

2.c) Autonomous Error Dynamics:

Question:

Verify numerically: Choose random $X, Y \in SE(2)$ and a twist $\xi = (15, 0, 0.3)^T$. Compute both sides of the group affine equation and confirm agreement to machine precision.

Solution:

Code snippet 1 shows the verification of group affine property of the $SE(2)$ unicycle dynamics

Code Snippet 1: Unicycle Group Affine Test

```

1 import pytest as pt
2 import numpy as np
3 from Code.SE2.SE2_maps import se2_compose, se2_inverse, se2_exp, se2_log, se2_vee, se2_wedge
  , se2_Ad, se2_ad
4
5 def test_se2_group_affine():
6     """
7     Test that group affine property holds for random SE2 elements with unicycle dynamics f_u
8     (X) = X xi^
9     f_u(XY) = f_u(X) Y + X f_u(Y) - X f_u(I) Y
10    """
11    n_samples = 100
12
13    rng = np.random.default_rng()
14    xi_X_random = np.reshape(rng.uniform(-10, 10, 3 * n_samples), (3, n_samples))
15    xi_Y_random = np.reshape(rng.uniform(-10, 10, 3 * n_samples), (3, n_samples))
16
17    xi = np.array([[15], [0], [0.3]])
18
19    checks = np.zeros([n_samples, 1])
20    for i in range(n_samples):
21        X_random = se2_exp(xi_X_random[:, i])
22        Y_random = se2_exp(xi_Y_random[:, i])
23
24        lhs = X_random @ Y_random @ se2_wedge(xi) # f_u(XY)
25        rhs = X_random @ se2_wedge(xi) @ Y_random + X_random @ Y_random @ se2_wedge(xi) -
26        X_random @ se2_wedge(xi) @ Y_random # f_u(X) Y + X f_u(Y) - X f_u(I) Y
27
28        checks[i] = np.all(lhs == pt.approx(rhs))
29
30    assert np.all(checks)

```

Figure 1: Group Affine Unicycle Test Passing

```

(.venv) PS C:\Users\thatf\OneDrive\Documents\Purdue Classes\AAE 590LGM\AAE590LGM-Lie-Group-Methods> pytest Tests\SE2\test_SE2_group_affine.py
===== test session starts =====
platform win32 -- Python 3.10.11, pytest-9.0.2, pluggy-1.6.0
rootdir: C:\Users\thatf\OneDrive\Documents\Purdue Classes\AAE 590LGM\AAE590LGM-Lie-Group-Methods
collected 1 item

Tests\SE2\test_SE2_group_affine.py . [100%]

===== 1 passed in 0.16s =====
(.venv) PS C:\Users\thatf\OneDrive\Documents\Purdue Classes\AAE 590LGM\AAE590LGM-Lie-Group-Methods>

```